MATH 562: STATISTICAL INFERENCE

Homework 3

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Contents

1 Preliminaries

We hope to introduce the reader to the main ideas and methods that will be used throughout this work.

Definition 1.1 (Chain rule for densities)**.** For random variables $Y_1, \dots, Y_n$,

$$f_{Y_1,\dots,Y_n}(y_1, \dots, y_n) = f_{Y_1}(y_1) \prod_{j=2}^{n} f_{Y_j|Y_1,\dots,Y_{j-1}}(y_j \mid y_1, \dots, y_{j-1}).$$

This identity, called the *prediction decomposition*, expresses the joint density as the product of one-step-ahead conditional densities.

1.1 AR(1) Gaussian Process

The following process is an autoregressive model of order 1 (AR(1)):

$$Y_j = \mu + \alpha(Y_{j-1} - \mu) + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma^2),$$

whuch satisfies the following conditional distribution:

$$Y_j \mid Y_{j-1} = y_{j-1} \sim \mathcal{N}\{\mu + \alpha(y_{j-1} - \mu),\, \sigma^2\}.$$

1.2 Likelihood

As a preliminary note, we define the likelihood function $L(\mathbf{y}|\theta)$ as $\mathbb{P}(\text{seeing data}|\theta) = \mathbb{P}(y_1, \dots, y_n|\theta)$ which 'splits' due to the property of independence. More formally,

Definition 1.2 (Likelihood)**.** Let $\mathbf{y} = (y_1, \dots, y_n)$ be i.i.d. samples from a probability mass function (pmf) $p_Y(t|\theta)$ (if Y is discrete) or probability density function (pdf) $f_Y(t|\theta)$ (if Y is continuous) parameterized (with parameter) by $\theta \in \Theta$. We define the likelihood of $\mathbf{y}$ given θ to be the "probability" of observing $\mathbf{y}$ if the true parameter is θ. If Y is discrete,

$$L(\mathbf{y}|\theta) = \prod_{i=1}^{n} p_Y(y_i|\theta)$$

and if Y is continuous,

$$L(\mathbf{y}|\theta) = \prod_{i=1}^{n} f_Y(y_i|\theta).$$

Hence, the likelihood function for data $y_0, \dots, y_n$ from an AR(1) Gaussian process with known initial value y_0 is given by

$$L(\mu, \alpha, \sigma^2) = \prod_{j=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}[y_j - \mu - \alpha(y_{j-1} - \mu)]^2\right\}.$$

Remark 1.3. The log-likelihood function is often more convenient to work with, as it transforms products into sums. The log-likelihood for the AR(1) Gaussian process is

$$\ell(\mu, \alpha, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^{n} [y_j - \mu - \alpha(y_{j-1} - \mu)]^2.$$

2 Exercise

2.1 Defining the problem

Observations $Y_1, \dots, Y_n$ arise in time order.

1. Starting from

$$f_{Y_1,\dots,Y_n}(y_1,\dots,y_n) \;=\; f_{Y_n|Y_1,\dots,Y_{n-1}}(y_n|y_1,\dots,y_{n-1})f_{Y_1,\dots,Y_{n-1}}(y_1,\dots,y_{n-1}),$$

   establish the *prediction decomposition*

$$f_{Y_1,\dots,Y_n}(y_1,\dots,y_n) = f_{Y_1}(y_1)\prod_{j=2}^{n} f_{Y_j|Y_1,\dots,Y_{j-1}}(y_j|y_1,\dots,y_{j-1}).$$

2. A stationary first-order Gaussian autoregressive process satisfies

$$Y_j|Y_1 = y_1,\dots,Y_j|Y_{j-1} = y_{j-1} \sim \mathcal{N}(\mu + \alpha(y_{j-1}-\mu),\sigma^2) \text{ for } j = 1,\dots,n$$

   with $|\alpha| < 1$, $\mu \in \mathbb{R}$, and $\sigma^2 > 0$. Find the log likelihood for data $y_0,\dots,y_n$ for this model if the initial value y_0 is treated

   (a) fixed as known constant,

   (b) coming from the stationary distribution $\mathcal{N}(\mu, \frac{\sigma^2}{1-\alpha^2})$

3. Give a minimal sufficient statistic for $\theta = (\mu,\sigma^2,\alpha)$ in (2). Does the model with parameter vector θ form an exponential family?

2.2 Solution

We first show the prediction decomposition formula starting from the given identity on (1).

$$f_{Y_1,\dots,Y_n}(y_1,\dots,y_n) \;=\; f_{Y_n|Y_1,\dots,Y_{n-1}}(y_n|y_1,\dots,y_{n-1})f_{Y_1,\dots,Y_{n-1}}(y_1,\dots,y_{n-1})$$

$$= f_{Y_n|Y_1,\dots,Y_{n-1}}(y_n|y_1,\dots,y_{n-1})f_{Y_{n-1}|Y_1,\dots,Y_{n-2}}(y_{n-1}|y_1,\dots,y_{n-2})f_{Y_1,\dots,Y_{n-2}}(y_1,\dots,y_{n-2})$$

$$= f_{Y_n|Y_1,\dots,Y_{n-1}}(y_n|y_1,\dots,y_{n-1})f_{Y_{n-1}|Y_1,\dots,Y_{n-2}}(y_{n-1}|y_1,\dots,y_{n-2})\cdots f_{Y_2|Y_1}(y_2|y_1)f_{Y_1}(y_1)$$

$$= f_{Y_1}(y_1)\prod_{j=2}^{n} f_{Y_j|Y_1,\dots,Y_{j-1}}(y_j|y_1,\dots,y_{j-1}).$$

Now, suppose we observe data $y_1,\dots,y_n$ from the autoregressive process

$$Y_j = \mu + \alpha(Y_{j-1}-\mu) + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0,\sigma^2),$$

with $|\alpha| < 1$. The main (small but important) difference in writing the likelihood starts from how we treat the initial condition Y_0:

(a) Conditional (profile) likelihood. If the initial value y_0 is treated as a fixed, known constant, we condition on it and write

$$L_c(\mu, \alpha, \sigma^2\,;\, y_1, \ldots, y_n \mid y_0) = \prod_{j=1}^{n} f(y_j \mid y_{j-1};\, \mu, \alpha, \sigma^2),$$

where

$$Y_j \mid Y_{j-1} = y_{j-1} \sim \mathcal{N}\big(\mu + \alpha(y_{j-1} - \mu),\, \sigma^2\big).$$

Hence, the conditional log-likelihood is

$$\ell_c(\mu, \alpha, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{j=1}^{n}[y_j - \mu - \alpha(y_{j-1} - \mu)]^2.$$

Here y_0 enters only in the first term (i.e. the term $y_1 \mid y_0$), but no density is assigned to y_0 by itself.

(b) Exact (unconditional) likelihood under stationarity. If we assume the process is stationary, then

$$Y_0 \sim \mathcal{N}\left(\mu,\, \frac{\sigma^2}{1-\alpha^2}\right),$$

and so, weget that the marginal (stationary) distribution of the first observation is

$$Y_1 \sim \mathcal{N}\left(\mu,\, \frac{\sigma^2}{1-\alpha^2}\right).$$

Integrating out the unobserved Y_0 leads to the *exact likelihood*

$$L_e(\mu, \alpha, \sigma^2\,;\, y_1, \ldots, y_n) = f\Big(y_1; \mu, \tfrac{\sigma^2}{1-\alpha^2}\Big)\prod_{j=2}^{n} f(y_j \mid y_{j-1};\, \mu, \alpha, \sigma^2),$$

and the corresponding log-likelihood is

$$\ell_e(\mu, \alpha, \sigma^2) = -\frac{1}{2}\log\frac{2\pi\sigma^2}{1-\alpha^2} - \frac{(y_1-\mu)^2(1-\alpha^2)}{2\sigma^2} - \frac{n-1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{j=2}^{n}[y_j - \mu - \alpha(y_{j-1} - \mu)]^2.$$

Interpretation:

- The **conditional likelihood** treats y_0 as a fixed starting point and models only the transitions $Y_j \mid Y_{j-1}$ for $j = 1, \ldots, n$.
- The **exact likelihood** assumes Y_0 (and hence Y_1) come from the stationary distribution and therefore includes their marginal densities.
- Both forms give consistent estimates of (μ, α, σ^2) as $n \to \infty$, but they differ slightly in small samples.

Minimal Sufficient Statistic

Note that likelihood depends on the data only through (if expanded and rearranged, the term in the summation that is)

$$\hat{T}(y) = \left(\sum_{j=1}^{n} y_j, \sum_{j=1}^{n} y_{j-1}, \sum_{j=1}^{n} y_j y_{j-1}, \sum_{j=1}^{n} y_j^2, \sum_{j=1}^{n} y_{j-1}^2 \right),$$

Remark 2.1. Note that the term $\sum_{j=1}^{n} y_{j-1}^2$ can be omitted since it can be computed from $\sum_{j=1}^{n} y_j^2$ and the known initial value y_0 as

$$\sum_{j=1}^{n} y_{j-1}^2 = \sum_{j=0}^{n-1} y_j^2 = \sum_{j=1}^{n} y_j^2 + y_0^2 - y_n^2.$$

and so, we define $T(y) = \left(\sum_{j=1}^{n} y_j, \sum_{j=1}^{n} y_{j-1}, \sum_{j=1}^{n} y_j y_{j-1}, \sum_{j=1}^{n} y_j^2 \right)$ which is then minimal sufficient for $\theta = (\mu, \sigma^2, \alpha)$.

Exponential Family

Since the Gaussian family is an exponential family, so is the AR(1) model. Hence we conclude the model with parameter vector θ forms an exponential family and thus the work is done.